

A PROJECT REPORT ON

**Blockchain-Based Intellectual Property and Will
Storage with OCR Automation**

SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING (Computer Engineering)

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2024-2025



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ACKNOWLEDGEMENT

*It gives us great pleasure in presenting the project report on ‘**Blockchain-Based Intellectual Property and Will Storage with OCR Automation**’.*

*We would like to take this opportunity to thank **Prof. P. S. Shahane** giving me all the help and guidance we needed. We are really grateful to them for their kind support. Their valuable suggestions were very helpful.*

*We are also grateful to **Dr. G. V. Kale**, Head of Computer Engineering Department, PICT for his indispensable support, suggestions.*

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ABSTRACT

This project presents a blockchain-based solution for the secure, decentralized storage and verification of Intellectual Property (IP) and Will documents. The system integrates the Ethereum blockchain for immutable, timestamped records and the InterPlanetary File System (IPFS) for distributed file storage, ensuring tamper-proof custody and verifiability without reliance on centralized authorities. To enhance usability and automation, Optical Character Recognition (OCR) technology is employed for extracting and digitizing information from scanned or handwritten legal documents. Smart contracts written in Solidity automate IP registration, ownership verification, and Will management processes, while a customized Proof of Stake (PoS)-like consensus mechanism ensures energy efficiency, scalability, and cost-effectiveness. The proposed system addresses the limitations of traditional document storage methods and existing blockchain-based solutions by offering a robust, transparent, and sustainable framework for protecting legal ownership and preventing forgery or disputes. The platform demonstrates strong potential for application across creative, legal, and personal estate planning domains.

Keywords: Blockchain, Intellectual Property (IP), Will Documents, Ethereum, IPFS (InterPlanetary File System), Smart Contracts, Optical Character Recognition (OCR), Proof of Stake (PoS).

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CHAPTER 1

INTRODUCTION

1.1 Overview

In the digital age, safeguarding intellectual property (IP) and legal documents such as Wills has become a growing challenge. Centralized storage methods are vulnerable to tampering, unauthorized access, and data loss. This project introduces a web-based platform that leverages Ethereum blockchain, Inter-Planetary File System (IPFS), and Optical Character Recognition (OCR) to build a secure, transparent, and tamper-proof system for document storage and verification. By integrating decentralized technologies and smart contracts, the platform ensures immutable, timestamped proof of ownership and enables automated, efficient document handling without the need for intermediaries.

1.2 Motivation

Creators, inventors, and individuals often face legal disputes and theft of their IP due to a lack of credible, timestamped proof of ownership. Similarly, traditional Will documents stored physically or in centralized databases are prone to loss or manipulation. Existing solutions are costly, inefficient, and often energy-intensive due to reliance on outdated blockchain mechanisms. These challenges inspired the development of a sustainable, automated, and scalable solution that empowers users to secure their legal records without needing technical expertise or relying on third parties.

1.3 Problem Definition and Objectives

In today's digital ecosystem, safeguarding intellectual property (IP) and legal documents like Wills remains a significant challenge due to the absence of tamper-proof, verifiable ownership mechanisms. Traditional IP protection methods, including copyright registrations and physical document storage, are often centralized, susceptible to loss, manipulation, and lack of accessibility. Moreover, the manual verification of these documents is time-consuming and prone to human error. Existing blockchain-based solutions, while promising, are either energy-intensive due to Proof of Work (PoW) or too complex for general users. There is a clear need for a decentralized, energy-efficient, and user-friendly system that enables individuals to create, store, and verify their legal documents with assured immutability and transparency, without reliance on intermediaries.

The key objectives of the project are as follows:

- To develop a secure and decentralized platform for storing Intellectual Property (IP) and Will documents using Ethereum blockchain and IPFS.
- To implement Optical Character Recognition (OCR) for automated metadata extraction from scanned or handwritten documents.
- To design and deploy Ethereum smart contracts for tamper-proof IP and Will registration, verification, and retrieval.
- To ensure system scalability and sustainability using a Proof of Stake (PoS)-inspired consensus mechanism without cryptocurrency-based validation.
- To provide a user-friendly interface that simplifies document upload, access, and integrity verification for non-technical users.
- To address key limitations of traditional and existing blockchain-based solutions in terms of security, cost, energy consumption, and centralization.

1.4 Project Scope & Limitations

This project aims to build a robust, scalable, and decentralized platform for the secure storage and verification of Intellectual Property (IP) and Will documents. The system leverages the Ethereum blockchain to provide immutable, timestamped proof of document existence, while integrating the InterPlanetary File System (IPFS) for efficient off-chain file storage. To streamline the process and reduce human error, the platform incorporates Optical Character Recognition (OCR) for automated extraction of metadata from scanned documents. Users can upload, verify, and retrieve documents through a user-friendly interface, with backend logic managing secure authentication and blockchain interactions. The solution is designed to serve a wide array of stakeholders—ranging from artists, innovators, and legal professionals to individuals seeking secure estate planning—offering transparency, auditability, and resilience against document tampering. This system represents a future-forward approach to digital rights management and legal documentation by eliminating centralized dependencies and ensuring long-term document integrity.

While the proposed system addresses many challenges, it does have certain limitations:

- The platform depends on Ethereum gas fees, which can fluctuate based on network congestion, potentially affecting cost efficiency.
- IPFS retrieval speed is dependent on the availability and responsiveness of network nodes, which may lead to occasional delays.
- The accuracy of the OCR module can degrade when handling low-resolution, noisy, or poorly formatted documents.
- Legal acceptance of blockchain-stored documents varies across jurisdictions, which may limit the platform's use in official legal proceedings.
- Adoption by non-technical users may require additional support or training due to complexities involved in blockchain interactions (e.g., wallet usage).
- As with all cryptographic systems, future advances in quantum computing may pose a threat to the current security models used in the platform.

1.5 Methodologies of Problem solving

The methodology adopted in this project focuses on leveraging blockchain and decentralized storage to ensure secure, transparent, and tamper-proof management of intellectual property (IP) and Will documents. At the core of the solution is the Ethereum blockchain, chosen for its support for smart contracts and wide adoption. When a user uploads a document, the system generates a cryptographic hash using Keccak-256, which is stored immutably on the blockchain. This hash acts as a digital fingerprint of the document, ensuring any future alteration can be instantly detected. To manage large files efficiently and reduce on-chain storage costs, the actual document is uploaded to the InterPlanetary File System (IPFS), a decentralized peer-to-peer file storage protocol. IPFS returns a unique Content Identifier (CID) for the file, which is then anchored on the blockchain to establish proof of existence.

To streamline the document registration process and reduce the need for manual data entry, Optical Character Recognition (OCR) technology is integrated into the system. This OCR module extracts key metadata such as document title, type, author, and legal identifiers from scanned or

handwritten documents. The extracted text is pre-verified against expected formats before being passed on to the blockchain. This significantly enhances the efficiency and accuracy of the platform, enabling a smooth user experience even for non-technical individuals. The OCR module is especially useful for handling legacy legal documents or records that were originally paper-based.

A critical component of the system is the set of smart contracts written in Solidity and deployed on Ethereum's Sepolia testnet. These contracts are responsible for automating document registration, ownership validation, and retrieval. Each transaction is time-stamped and linked to the user's credentials, providing a transparent and traceable audit trail. Functions such as `addIpRecord` and `addWillRecord` ensure that users can register their documents with detailed metadata, while `readIpRecord` and `readWillRecord` provide verifiable access to stored information. This automation eliminates the risk of human error, ensures consistency, and provides legally binding proof of record submission.

To maintain scalability and sustainability, a Proof of Stake (PoS)-inspired consensus mechanism is used. Unlike traditional PoW models, this approach avoids energy-intensive mining and reduces operational costs. While the platform does not rely on cryptocurrency-based staking, validators can be determined through non-monetary reputation or authorization models. This ensures efficient transaction validation with minimal resource usage, making the system suitable for large-scale deployment. Overall, the combined methodology of blockchain immutability, IPFS-based storage, OCR automation, and energy-efficient consensus results in a highly secure and future-ready document management framework.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Review

2.1.1 Blockchain-Based Traceability and Management for Additive Manufacturing

This paper discusses the use of blockchain for tracking and managing additive manufacturing processes. It highlights the challenges with Proof of Work (PoW) consensus, particularly its high energy usage and low scalability. The authors propose solutions like sharding and Layer-2 protocols to improve performance and support large-scale data handling. This directly supports the decision to use a PoS-based approach in our system.

2.1.2 A Proxy Re-Encryption Approach to Secure Data Sharing in IoT Based on Blockchain

Description: Focuses on secure data sharing using private blockchains and proxy re-encryption (PRE). It ensures data privacy in IoT systems. While not directly targeting IP management, the security model using PRE is relevant and inspires stronger encryption mechanisms in our system.

2.1.3 TA-SPESC: Asset-Driven Smart Contract Language on Blockchain

Introduces a smart contract language to support automated asset transactions on Ethereum. It enables ownership verification through rule-based conversions into Solidity contracts. This informs the design of our smart contracts for document ownership and management.

2.1.4 Securing IoT Devices with Blockchain

Highlights the use of blockchain to ensure data integrity in IoT environments. Although based on PoW, which incurs high energy costs, the work emphasizes the importance of decentralization and security—principles which our IP platform builds upon using more efficient PoS.

2.1.5 A Blockchain-Based Traceable IP Copyright Protection Algorithm

Proposes a copyright protection system using blockchain and traceable algorithms. It uses cryptographic techniques like quadratic matrix transforma-

tion to ensure document authenticity. This supports the use of hash-based verification in our solution.

2.1.6 Blockchain Mutability: Challenges and Proposed Solutions

Discusses issues with data immutability in blockchain systems and suggests solutions like sidechains and off-chain storage. This supports our integration of IPFS and awareness of blockchain's limitations in large-scale document handling.

2.2 Existing methodologies

2.2.1 Proof of Work (PoW) for Tamper-Proof Record Keeping

Several systems, like those discussed by Xiao et al. and Kim et al., rely on PoW to achieve immutability and traceability. While secure, PoW leads to high energy consumption and slow transaction processing, making it inefficient for large-scale IP document management.

2.2.2 Smart Contracts on Ethereum for Ownership and Automation

Works like TA-SPESC by Fran Casino et al. leverage Ethereum smart contracts to automate ownership verification and asset transactions. These contracts are crucial in ensuring that records remain immutable and verifiable without intermediaries.

2.2.3 Proxy Re-Encryption and Advanced Cryptography for Privacy

Studies such as Opuni-Boachie et al. integrate Proxy Re-Encryption (PRE) and cryptographic protocols to ensure secure data sharing and confidentiality. These are particularly valuable in managing sensitive documents like wills or legal records.

However, these above mentioned methods often struggle with scalability, cross-language support, and real-time adaptability, highlighting the need for more flexible and intelligent solutions.

2.3 Research Gap Analysis

After going through reference papers and material we identified following drawbacks in existing solutions:

1. **High Energy Consumption in Existing Platforms:** Many current IP protection systems, like Bernstein.io and Proof of Existence, still rely on Proof of Work (PoW), which consumes significant energy and is not environmentally sustainable for large-scale adoption.
2. **Lack of Automation in Document Processing:** Existing solutions often require manual entry and verification of document data. The absence of OCR integration leads to a time-consuming and error-prone process, especially when handling physical or scanned documents.
3. **Limited Scalability and High Costs:** Platforms built on PoW or without optimization techniques struggle with scalability and incur high transaction fees, making them impractical for widespread use across industries dealing with large volumes of data.
4. **Inadequate Data Privacy and User Control:** While blockchain ensures immutability, many systems lack granular access control, such as proxy re-encryption or user-defined privacy levels, limiting their utility for sensitive documents like wills.
5. **Low Legal and Public Acceptance:** Legal recognition of blockchain-stored documents is still evolving. User unfamiliarity with decentralized tools (e.g., MetaMask) and jurisdictional ambiguity affect adoption and trust in real-world legal scenarios.

2.4 Scope of Improvement

The current blockchain-based IP management systems provide a solid foundation for secure and tamper-proof document storage, but they still face several limitations. One major area for improvement is automation. Integrating technologies like Optical Character Recognition (OCR) can streamline the document registration process by automatically extracting text from scanned legal documents, reducing manual errors and saving time. This makes the system more accessible to users who are not technically inclined and improves data consistency across the platform.

Another key area is cost and performance optimization. Many existing platforms still depend on energy-intensive consensus mechanisms like Proof of Work (PoW), which leads to high operational costs and slower transaction times. Transitioning to Proof of Stake (PoS) or a customized PoS-like mechanism without cryptocurrency can drastically reduce energy consumption and improve scalability. Additionally, incorporating layer-2 scaling solutions and smart contract enhancements can ensure better throughput, lower gas fees, and wider adoption across sectors like legal services, estate planning, and creative industries.

CHAPTER 3

**SOFTWARE REQUIREMENTS
SPECIFICATION**

3.1 Assumptions and Dependencies

- Users will have access to the internet and basic computing devices.
- Ethereum blockchain and IPFS nodes are assumed to be active and accessible.
- Users are responsible for managing their private keys securely.
- Node.js and MongoDB are assumed to be properly configured on the backend server.

3.2 Functional Requirements

3.2.1 User Registration and Authentication

- The system shall allow users to register and log in using secure credentials.

3.2.2 Document Upload and Storage

- Users can upload intellectual property documents or wills.
- Documents are stored on IPFS and the hash is recorded on the Ethereum blockchain.

3.2.3 Verification and Retrieval

- The system enables users to verify and retrieve uploaded documents.
- Hash comparisons will validate the integrity of documents.

3.3 External Interface Requirements

3.3.1 User Interfaces

- A React-based web interface for uploading, verifying, and managing documents.
- Displays transaction status, document details, and verification results.

3.3.2 Hardware Interfaces

- No special hardware interfaces are required beyond a standard web-enabled device.

3.3.3 Software Interfaces

- Integration with Ethereum blockchain via Web3.js.
- IPFS integration for decentralized file storage.

3.3.4 Communication Interfaces

- HTTPS for all user-platform interactions.
- Web3 provider (e.g., Infura) for blockchain interactions.

3.3.5 Software Interfaces

1. Operating System: Anyone among MAC OSX, Linux or Windows
2. Jupyter Notebook
3. Dataset pertinent to medical images
4. Programming Language: Python

3.3.6 Communication Interfaces

3.4 System Requirements

3.4.1 Database Requirements

- MongoDB to store metadata (user details, document info, timestamps).

3.4.2 Software Requirements (Platform Choice)

- Backend: Node.js with Express.js
- Frontend: React.js
- Blockchain: Ethereum with smart contracts in Solidity
- Storage: IPFS

3.4.3 Hardware Requirements

- Standard server with:
 - 8 GB RAM or higher
 - Quad-core CPU
 - 100 GB SSD storage

3.5 Analysis Models: SDLC Model to be Applied

- The **Agile Model** will be used to allow iterative development and continuous feedback.
- Regular sprints with testing and deployment phases will ensure timely updates and bug fixes.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The proposed system architecture integrates multiple decentralized technologies to ensure secure, transparent, and tamper-proof management of Intellectual Property (IP) and Will documents. Users interact with the platform through a web-based interface built using React.js, where they can register, authenticate, and upload documents. Upon upload, the files are processed using Optical Character Recognition (OCR) to extract critical information, which is then stored in a decentralized manner using the InterPlanetary File System (IPFS). A unique Content Identifier (CID) is generated for each file and recorded on the Ethereum blockchain through smart contracts written in Solidity. These contracts automate key functions such as document registration, verification, and ownership transfer. The backend, developed with Node.js and Express.js, handles user data and communicates with both IPFS and Ethereum via Web3.js. A PoS-inspired consensus mechanism ensures energy-efficient transaction validation, while a verification module cross-checks hashes between IPFS and the blockchain to confirm document integrity. This layered architecture ensures data immutability, scalability, and enhanced user trust without reliance on centralized authorities.

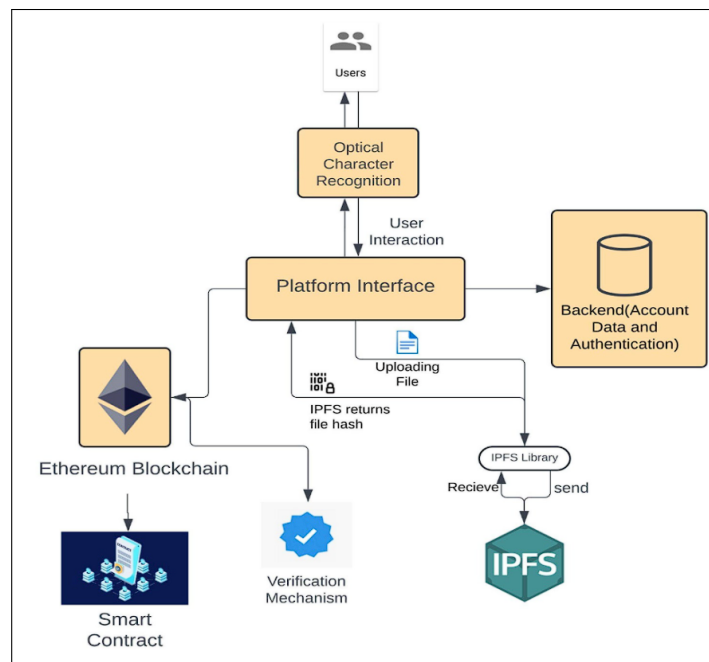


Figure 4.1: System Architecture

4.2 Data Flow Diagrams

- **User Interface (Frontend)**
 - The user accesses the platform via a React-based web interface.
 - User actions include registration/login, document upload, and verification requests.
- **Authentication**
 - Users authenticate using MetaMask (blockchain wallet).
 - Secure login tokens or session data are managed via backend logic.
- **Document Upload and OCR Processing**
 - The uploaded document is sent to the OCR engine for text extraction.
 - Extracted metadata (title, description, identifiers) is forwarded to the backend.
- **IPFS Integration (Decentralized File Storage)**
 - The original document is uploaded to the InterPlanetary File System (IPFS).
 - IPFS returns a unique Content Identifier (CID), acting as the document fingerprint.
- **Blockchain Interaction (Ethereum)**
 - The CID and relevant metadata are recorded on the Ethereum blockchain using smart contracts.
 - This ensures immutability and timestamping for proof of existence and ownership.
- **Database Storage (MongoDB)**
 - User data, document metadata, and blockchain transaction references are stored securely.
 - Used for quick access, dashboard views, and management operations.

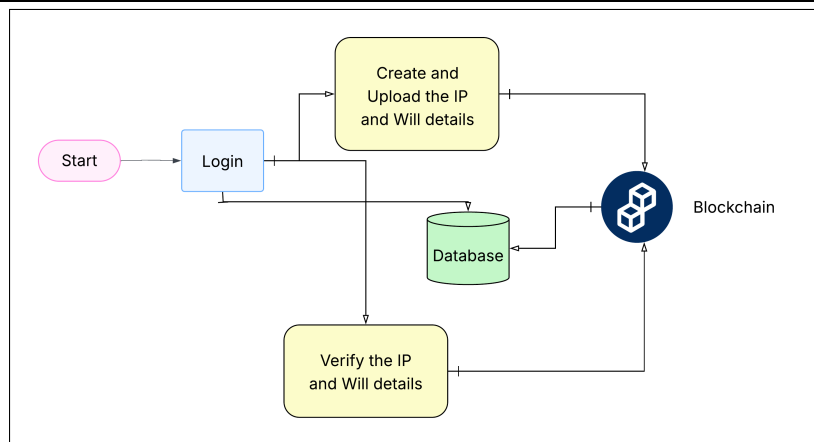


Figure 4.2: High Level Data Flow Diagram

4.3 Entity Relationship Diagram

MongoDB Schema Design

- **Users Collection**

- Stores basic user metadata such as **name**, **email**, **phone**, and **password**.
- Each user has two array references:
 - * **userIP**: links to their Intellectual Property documents.
 - * **userWill**: links to their Will documents.
- **Relationship**: One user can have **many** IPs and **many** Wills (1:M).

- ◇ **UserIP Collection**

- Stores IP document metadata:
 - * **type**, **title**, **blockNumber**, **txnhash**, and **documentId**.
- These are linked to a user via **userIP** array in the Users collection.

- ◇ **UserWill Collection**

- Stores Will document metadata:
 - * Fields are similar to UserIP: **type**, **title**, **blockNumber**, **txnhash**, and **documentId**.
- Linked to the user through the **userWill** array.

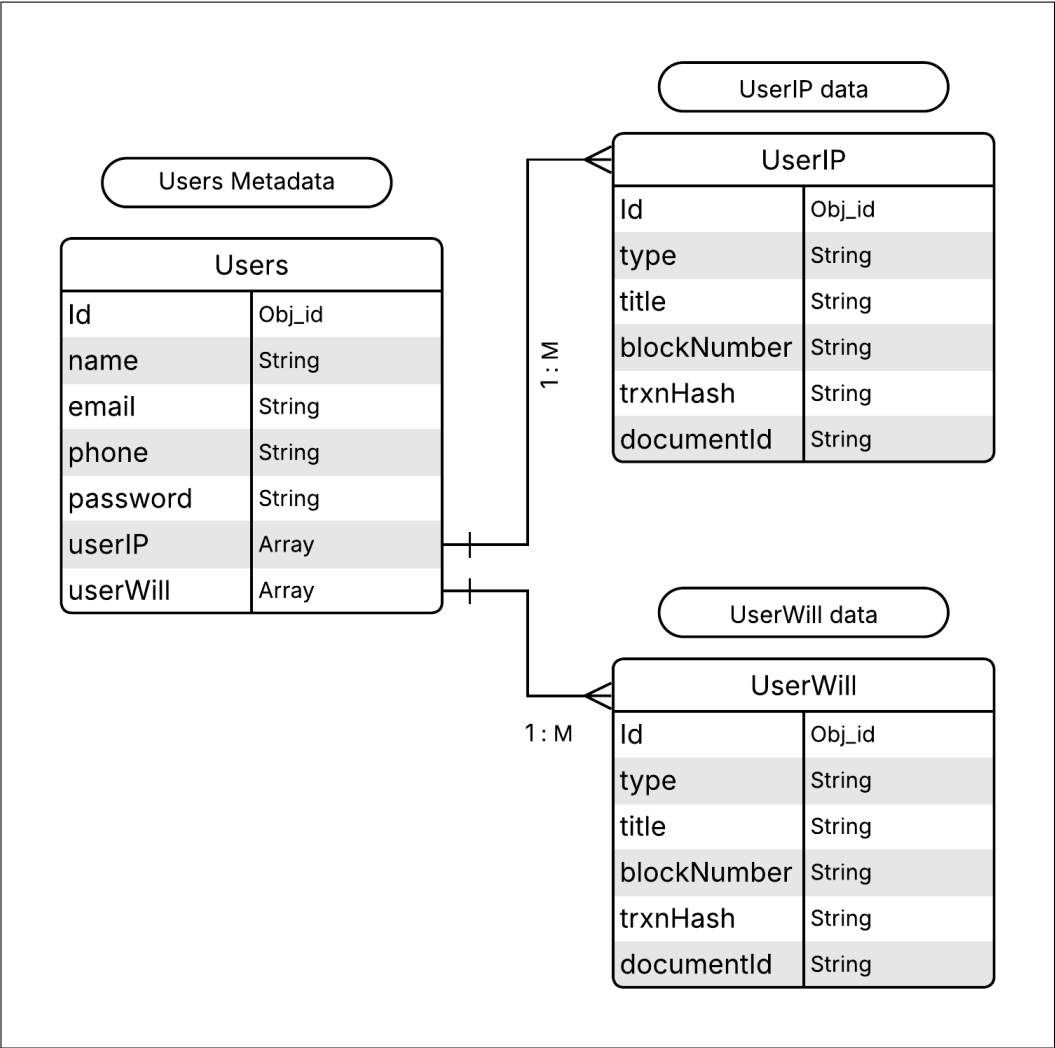


Figure 4.3: Entity Relationship Diagram

4.4 UML Diagram

This UML diagram represents the NoSQL database structure for a blockchain-based system that manages Intellectual Property (IP) and Will documents. The central collection is Users, which stores basic user information such as name, email, phone number, and password. Each user document includes two arrays: userIP and userWill, representing one-to-many relationships with the UserIP and UserWill collections, respectively. The UserIP and UserWill collections store document-specific metadata including type, title, blockchain transaction hash (txnHash), block number, and a unique documentId linked to IPFS. This design enables efficient and scalable document management and verification using decentralized technologies, ideal for MongoDB or similar NoSQL databases.

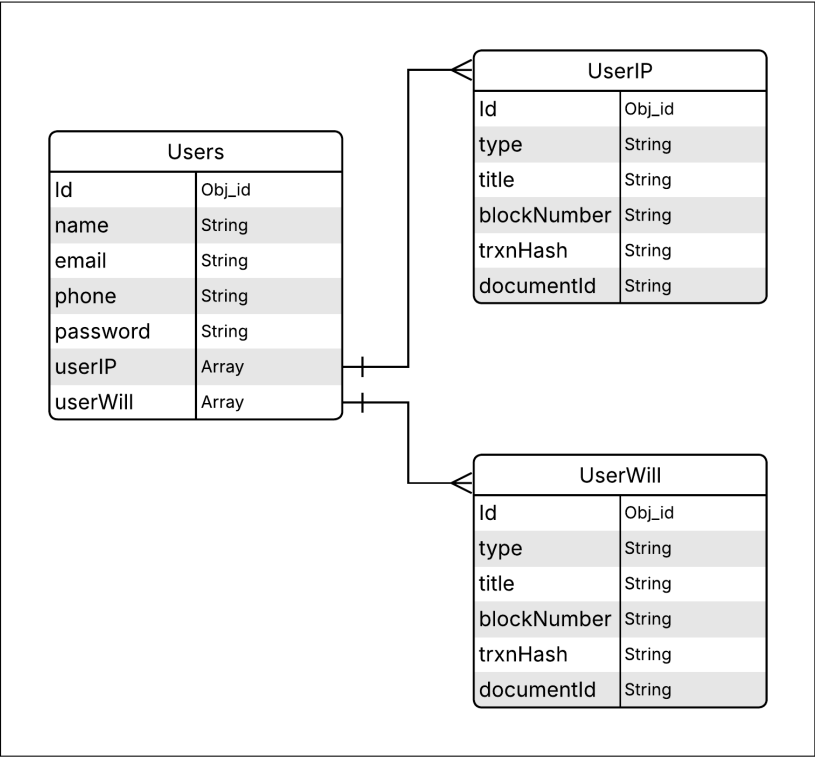


Figure 4.4: UML Diagram

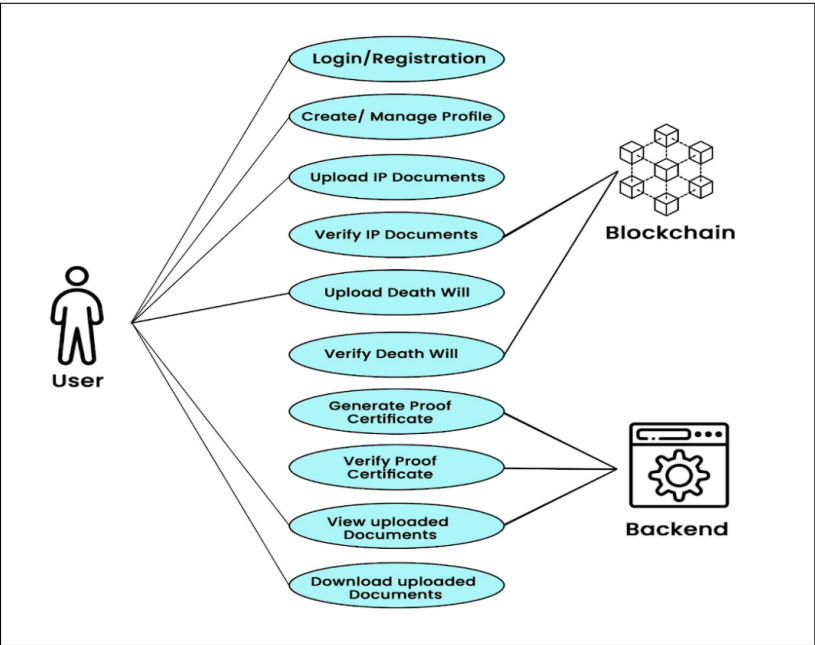


Figure 4.5: Use Case Diagram 1

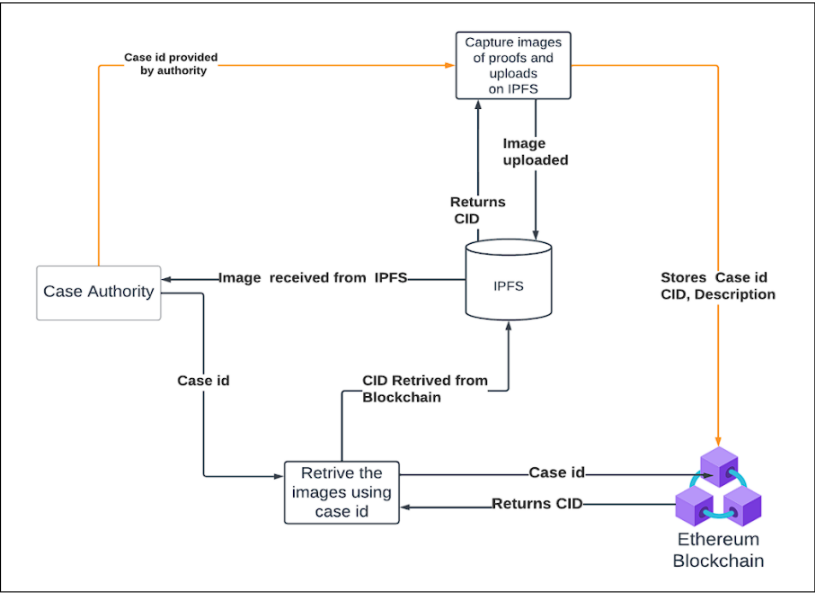


Figure 4.6: Use Case Diagram 2

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

- **Development Time:** Estimated at 4 months including testing and deployment.
- **Man-Hours:** ~800 cumulative hours across the team.
- **Budget:** INR 25,000 including hosting, APIs, and miscellaneous expenses.

5.1.2 Project Resources

- **Tools:** Visual Studio Code, MetaMask, Ganache, MongoDB Atlas, IPFS Desktop.
- **Technologies:** Ethereum (Solidity), IPFS, Node.js, Express.js, React.js.
- **Platforms:** Sepolia Testnet (for smart contract testing), GitHub for version control.

5.2 Risk Management

5.2.1 Risk Identification

- **Technical Risk:** Bugs in smart contract logic leading to loss of data.
- **Operational Risk:** IPFS node unavailability affecting file access.
- **User Risk:** Low adoption due to unfamiliarity with blockchain wallets.

5.2.2 Risk Analysis

- **High:** Smart contract vulnerabilities, security breaches.
- **Medium:** Network downtime (IPFS/Ethereum).
- **Low:** UI/UX confusion during onboarding.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

- Use of **testnets** and code audits for smart contract reliability.
- **Redundant IPFS nodes** and gateway backups.
- Provide **training materials** for MetaMask and blockchain basics.

5.3 Project Schedule

5.3.1 Project Task Set

- **Phase 1:** Requirement Analysis & Design (Week 1–2)
- **Phase 2:** Frontend and Backend Development (Week 3–6)
- **Phase 3:** Smart Contract Development & IPFS Integration (Week 7–9)
- **Phase 4:** Testing and Deployment (Week 10–12)
- **Phase 5:** Final Report & Documentation (Week 13–14)

5.3.2 Task Network

- Tasks are organized in a **linear-iterative flow**, where smart contracts and backend APIs are tested in parallel with the frontend UI.

5.4 Team Organization

5.4.1 Team Structure

- **Darshan Mahajan** – Backend & Smart Contracts Lead
- **Gaurav Patil** – Documentation, Deployment & Support
- **Rahul Patil** – Frontend Development & UI Design
- **Pratik Patil** – IPFS Integration & Testing

5.4.2 Management Reporting and Communication

- Weekly team sync-ups for progress review and task allocation.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

The project is divided into several core modules to ensure secure, decentralized, and efficient management of intellectual property (IP) and will-related documents:

- **User Authentication Module:** Handles secure registration and login via MetaMask.
- **Document Upload Module:** Allows users to upload IP or will documents.
- **IPFS Integration Module:** Converts files to hashes (CIDs) and stores them on the IPFS network.
- **Blockchain Module:** Records the hash on Ethereum for tamper-proof verification.
- **Document Verification Module:** Enables users to validate document authenticity via hash comparison.
- **Admin Dashboard:** For monitoring user activities and managing platform health.

6.2 Tools and Technologies Used

- **Frontend:** React.js (User interface design)
- **Backend:** Node.js and Express.js (Server logic)
- **Blockchain:** Ethereum (Smart contract deployment via Solidity)
- **Decentralized Storage:** IPFS (File storage)
- **Database:** MongoDB (Metadata storage)
- **Testing Network:** Sepolia Ethereum Testnet
- **Other Tools:** Web3.js, Ganache, Postman, Visual Studio Code

6.3 Algorithm Details

6.3.1 IPFS Document Storage

1. Convert the uploaded document into binary format.
2. Pass it to the IPFS library to generate a unique Content Identifier (CID).
3. Return the CID and link it to the user's metadata in MongoDB.
4. Use this CID to retrieve the document from IPFS when needed.

6.3.2 Blockchain Hash Recording

1. Hash the document locally using Keccak-256.
2. Create a smart contract function (e.g., `storeHash(cid, timestamp)`).
3. Call the function from the frontend using Web3.js and MetaMask.
4. Record the transaction ID and block number for audit purposes.

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing

7.1.1 Unit Testing

Conducted on individual components such as smart contract functions, backend APIs, and frontend modules.

7.1.2 Integration Testing

Verified interactions between modules like frontend ↔ backend ↔ blockchain/IPFS. Ensured data flow correctness, e.g., document hash from frontend reaches Ethereum contract via backend.

7.1.3 System Testing

End-to-end functionality tested including user registration, document upload, verification, and retrieval. Verified seamless integration across all technologies (Ethereum, IPFS, MongoDB, etc.).

7.1.4 Usability Testing

Evaluated ease of use, particularly for non-technical users dealing with MetaMask and document uploads. Improved UI elements based on user feedback.

7.2 Test cases & Test Results

Description	Expected Result	Actual Result	Status
User registration via MetaMask	Wallet connected & user data saved	Passed	Passed
Upload IP document to IPFS	CID generated successfully	CID returned & stored	Passed
Verify uploaded document	Hash matches with blockchain	Hash verified correctly	Passed
Unauthorized document access	Access denied	Access denied	Passed
Document retrieval after logout	Should prompt login	Prompted MetaMask login	Passed

Table 7.1: Test Cases and Results

CHAPTER 8

RESULTS

8.1 Outcomes

Table 8.1: Performance Metrics

Metrics	Ethereum + IPFS	Ethereum + OCR
Average Block Time (seconds)	~ 12 - 15	~ 10 - 12
IPFS Fetch Time (seconds)	~ 5 - 10	NA
OCR Processing Time (seconds)	NA	10 - 15 (varies with doc size)
End-to-End Transaction Time (seconds)	20 - 30	28 - 38

Table 8.2: Data Size and Storage Cost

Metrics	Ethereum + IPFS	Ethereum + OCR
Average Metadata Size (bytes)	~ 32	~ 1200
On-Chain Storage Cost (per kB)	~ \$0.002 - \$0.005	~ \$0.002 - \$0.005
Off-Chain Storage Cost (per MB)	~ \$0.01 - \$0.03	NA
Monthly Storage Costs (off-chain, GB)	~ \$10 - \$30	NA

Table 8.3: Data Integrity and Security

Metrics	Ethereum + IPFS	Ethereum + OCR
Hashing Algorithm	Keccak-256 (Ethereum)	Keccak-256 (Ethereum)
Data Tamper Resistance	High	Moderate
Document Verification Feasibility	Hash Comparison	Text and hash comparison
Security Level (for data storage)	Strong (IPFS)	Medium to Strong (OCR reliability)

Table 8.4: Scalability and System Performance

Metrics	Ethereum + IPFS	Ethereum + OCR
Scalability	High	High
OCR Processing Scalability	NA	Moderate (depends on server)
System Performance with Load	Moderate to High	Moderate (OCR load-dependent)
Maximum Supported TPS	12 - 15	10 - 12

8.2 Screen Shots

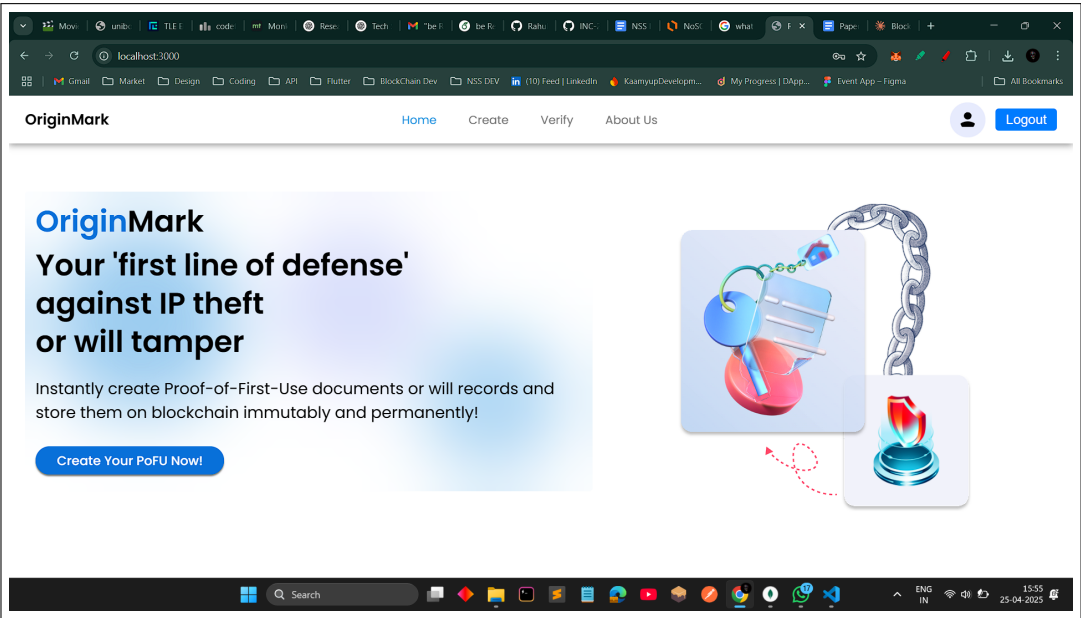


Figure 8.1: Home Page

Blockchain-Based Intellectual Property and Will Storage with OCR Automation

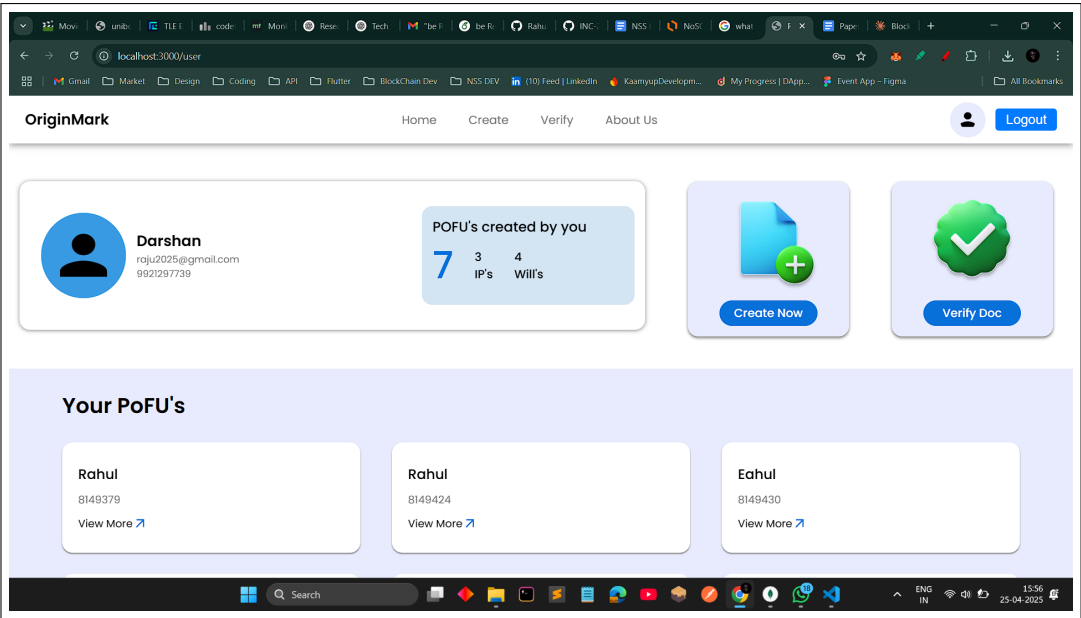


Figure 8.2: User Profile and PoFU's Data

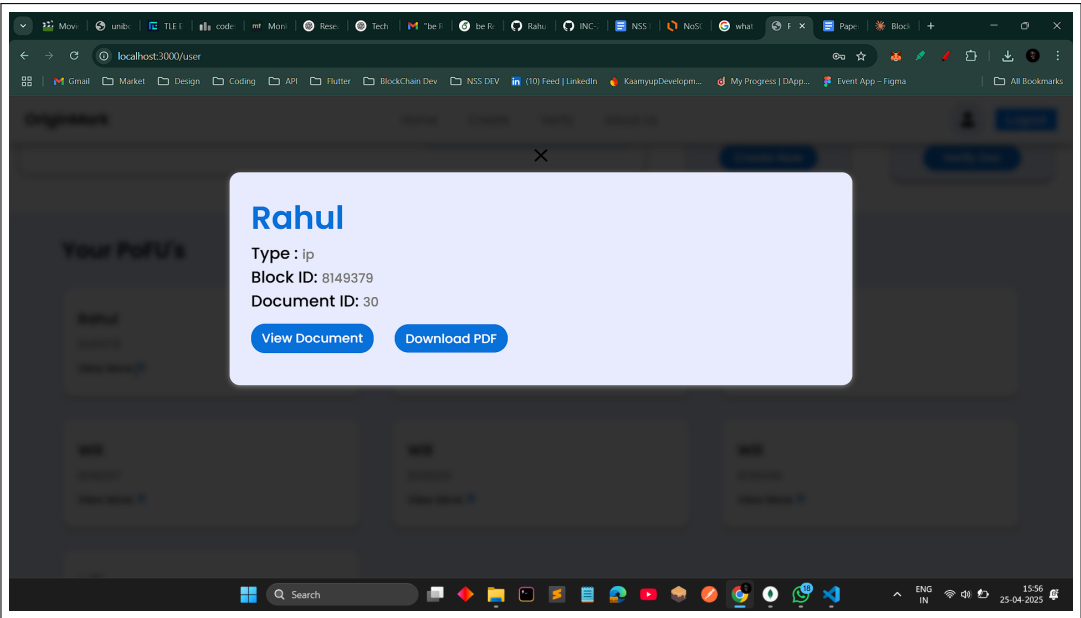


Figure 8.3: IP Details

CHAPTER 9

CONCLUSIONS

9.1 Conclusion 1

Secure and Decentralized Document Management: Our platform uses Ethereum blockchain and IPFS technologies to provide a secure, tamper-proof, and decentralized system for managing intellectual property (IP) and will documents, ensuring immutability, transparency, and protection against unauthorized access.

9.2 Conclusion 2

Automation and Trust through Smart Contracts: Smart contracts automate critical processes like document verification and ownership transfer, eliminating intermediaries and enhancing the system's trustworthiness and efficiency.

9.3 Conclusion 3

Empowering Users and Advancing Decentralized Solutions: By giving users full control over their sensitive documents and offering a transparent, efficient management system, our project addresses key security challenges and represents a major advancement in decentralized document management.

9.4 Future Work

- **Multi-Blockchain Support:** Expand the platform to support multiple blockchain networks such as Binance Smart Chain and Polygon, offering greater flexibility, lower transaction costs, and improved scalability.
- **AI-Driven Automation:** Incorporate AI technologies like natural language processing (NLP) for document summarization and smart classification to efficiently manage large volumes of documents.
- **Legal System Integration:** Develop capabilities for interaction with legal systems and third-party services to enable automatic legal validation of documents across different jurisdictions.

- **Mobile-First Design Approach:** Prioritize a mobile-friendly interface to ensure seamless access and usability, especially in regions where mobile internet usage dominates.

9.5 Applications

- **Intellectual Property (IP) Protection:** Secure storage and verification of copyrights, patents, trademarks, and other intellectual properties, ensuring proof of ownership and preventing disputes.
- **Digital Will Management:** Safekeeping and automated execution of digital wills through smart contracts, enabling transparent and tamper-proof inheritance processes.
- **Legal Document Verification:** Authenticating legal contracts, agreements, and certificates with blockchain timestamps, making them easily verifiable and reducing forgery risks.
- **Cross-border Document Management:** Providing an internationally recognized and decentralized system for secure document validation across different jurisdictions.
- **NFT-based Asset Management:** Extending the platform to tokenize IP assets as NFTs (Non-Fungible Tokens), enabling ownership transfer and royalty management in digital marketplaces.

CHAPTER 10

APPENDIX

10.1 Appendix A

1. Satisfiability Analysis

The core verification process can be modeled as a Boolean Satisfiability (SAT) problem:

$$Satisfiable \text{ if } \exists h_i \text{ such that } Verify(h_i, Blockchain) = True$$

where h_i is the hash of document d_i .

- Document verification reduces to a simple check for the existence of a valid hash on the blockchain.
- Therefore, the problem is **Satisfiable** under standard blockchain and IPFS operations.

2. Complexity Analysis: P, NP, NP-Complete, NP-Hard

The complexity classification of key operations is as follows:

Operation	Complexity	Class
Hashing document	$O(n)$	P
Storing hash on blockchain	$O(1)$ (transaction time)	P
Verifying document authenticity	$O(1)$ (hash comparison)	P
Uploading document to IPFS	$O(n)$	P
Ownership transfer via smart contract	$O(1)$	P

- All operations involved are solvable in polynomial time.
- No NP-Complete or NP-Hard problems are encountered in the core workflow.

Thus, the system belongs to the **P complexity class** and is computationally feasible.

3. Modern Algebra Perspective

Using concepts from Group Theory and Ring Theory:

- Hash functions such as SHA-256 and Keccak-256 act as **homomorphisms**:

$$H : (M, +) \rightarrow (H, +)$$

where M is the message space and H is the hash space.

- Blockchain operations form **cyclic groups** with hash chaining ensuring immutability.
- Collision resistance and non-reversibility of hashes imply a non-isomorphism from H back to M .

These algebraic structures reinforce the security and integrity of the platform.

Final Verdict

- **Satisfiability:** The document verification problem is satisfiable.
- **Complexity:** All core operations are in P (polynomial time).
- **Algebraic Structure:** Blockchain and IPFS operations align with Group Theory and Ring Theory.
- **Feasibility:** The project is highly feasible under modern computational and mathematical models.

Summary

The blockchain-based document management system is computationally feasible (P-type), structurally robust (SAT-satisfiable), and algebraically consistent (supported by Group Theory).

10.2 Appendix B

1. **Name of Conference/Journal:** Journal of Emerging Technologies and Innovative Research (JETIR 2025)
2. **Link to the Paper:** <https://www.jetir.org/papers/JETIR2501309.pdf>
3. **Certificates:**



Figure 10.1: Certificate 1



Figure 10.2: Certificate 2



Figure 10.3: Certificate 3



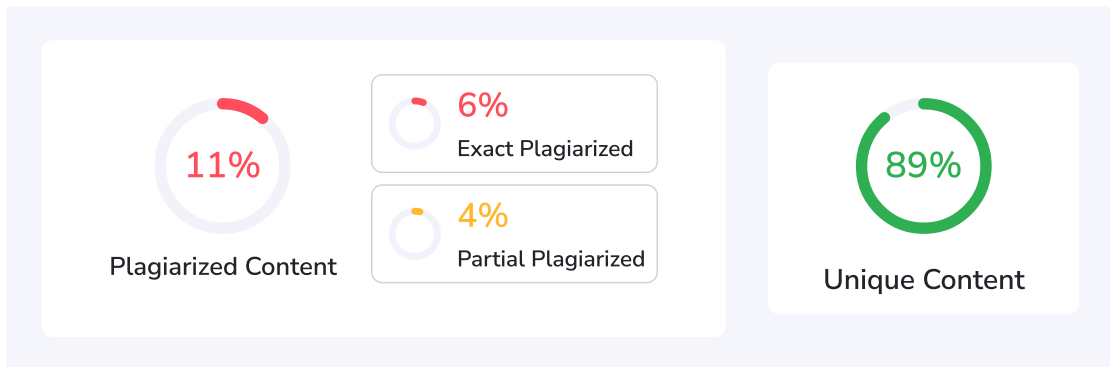
Figure 10.4: Certificate 4



Figure 10.5: Certificate 5

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A PROJECT REPORT ON Blockchain-Based Intellectual Property and Will Storage with OCR Automation SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF BACHELOR OF ENGINEERING (Computer Engineering) SUBMITTED BY Darshan Mahajan Exam No: B400050485 Gaurav Patil Exam No: B400050535 Pratik Patil Exam No: B400050539 Rahul Patil Exam No: B400050540 DEPARTMENT OF COMPUTER ENGINEERING Pune Institute of Computer Technology Dhankawadi, Pune - 411043 SAVITRIBAI PHULE PUNE UNIVERSITY 2024-2025 CERTIFICATE This is to certify that the project report entitled Blockchain-Based Intellectual Property and Will Storage with OCR Automation Submitted by Darshan Mahajan Exam No: B400050485 Gaurav Patil Exam No: B400050535 Pratik Patil Exam No: B400050539 Rahul Patil Exam No: B400050540 is a bonafide student of this institute and the work has been carried out by him/her under the supervision of Prof. P. S. Shahane and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune Uni- versity, for the award of the degree of Bachelor of Engineering (Computer Engineering). Prof. P. S. Shahane Dr. G. V. Kale Internal Guide Head Dept. of Computer Engg. Dept. of Computer Engg. Dr. P. T. Kulkarni Principal Pune Institute of Computer Technology Place:Pune Date: ACKNOWLEDGEMENT It gives us great pleasure in presenting the project report on 'Blockchain- Based Intellectual Property and Will Storage with OCR Automa- tion'. We would like to take this opportunity to thank Prof. P. S. Shahane giving me all the help and guidance we needed. We are really grateful to them for their kind support. Their valuable suggestions were very helpful. We are also grateful to Dr. G. V. Kale, Head of Computer Engineering Department, PICT for his indispensable support, suggestions. Darshan Mahajan Gaurav Patil Pratik Patil Rahul Patil (B.E. Computer Engg.) ABSTRACT This project presents a blockchain-based solution for the secure, decentralized storage and verification of Intellectual Property (IP) and Will documents. The system integrates the Ethereum blockchain for immutable, timestamped records and the InterPlanetary File System (IPFS) for distributed file storage, ensuring tamper-proof custody and verifiability without reliance on central- ized authorities. To enhance usability and automation, Optical Character Recognition (OCR) technology is employed for extracting and digitizing in- formation from scanned or handwritten legal documents. Smart contracts written in Solidity automate IP registration, ownership verification, and Will management processes, while a customized Proof of Stake (PoS)-like consen- sus mechanism ensures energy efficiency, scalability, and cost-effectiveness. The proposed system addresses the limitations of traditional document storage methods and existing blockchain-based solutions by offering a robust, trans- parent, and sustainable framework for protecting legal ownership and pre- venting forgery or disputes. The platform demonstrates strong potential for application across creative, legal, and personal estate planning domains. Keywords: Blockchain, Intellectual Property (IP), Will Documents, Ethereum, IPFS (InterPlanetary File System), Smart Contracts, Optical Character Recog- nition (OCR), Proof of Stake

(PoS). Contents	1	1.1 Overview	2	1.2 Motivation	2
	2	1.3 Problem Definition and Objectives	2	1.4 Project Scope & Limitations	3
	3	1.5 Methodologies of Problem solving	4	2 Literature Survey	6
	6	2.1 Literature Review	7	2.1.1 Blockchain-Based Traceability and Management for Additive Manufacturing	7
	7	2.1.2 A Proxy Re-Encryption Approach to Secure Data Sharing in IoT Based on Blockchain	7	2.1.3 TASPES: Asset-Driven Smart Contract Language on Blockchain	7
	7	2.1.4 Securing IoT Devices with Blockchain	7	2.1.5 A Blockchain-Based Traceable IP Copyright Protection Algorithm	7
	7	2.1.6 Blockchain Mutability: Challenges and Proposed Solutions	8	2.2 Existing methodologies	9
	9	2.2.1 Proof of Work (PoW) for Tamper-Proof Record Keeping	9	2.2.2 Smart Contracts on Ethereum for Ownership and Automation	9
	9	2.2.3 Proxy Re-Encryption and Advanced Cryptography for Privacy	9	2.3 Research Gap Analysis	10
	10	2.4 Scope of Improvement			

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We would like to take this opportunity to thank Prof. [↗](#)

https://www.ibo.org/globalassets/new-structure/research/pdfs/indiapypreport_final.pdf

is a bonafide student of this institute and the work has been carried out by him/her under the supervision of Prof. [↗](#)

<https://www.coursehero.com/file/217656829/BE-project-report-format-23-24pdf>

Darshan Mahajan Gaurav Patil Pratik Patil Rahul Patil (B.E. [↗](#)

https://www.linkedin.com/posts/darshan-mahajan-74179a23b_blockchain-innovation-pict-activity-7183078969016025088--1eY

We are really grateful to them for their kind support. [↗](#)

https://www.academia.edu/86020287/A_SURVEY_ON_SMART_ENERGY_METER_WITH_BILLING_AND_THEFT_DETECTION

To enhance usability and automation, Optical Character Recognition (OCR) technology is employed for extracting and digitizing information from scanned or handwritten legal documents. [↗](#)

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CHAPTER 11

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
IPFS	Interplanetary File System
IP	Intellectual Property
OCR	Optical Character Recognition
PoS	Proof Of Stake
PoW	Proof Of Work